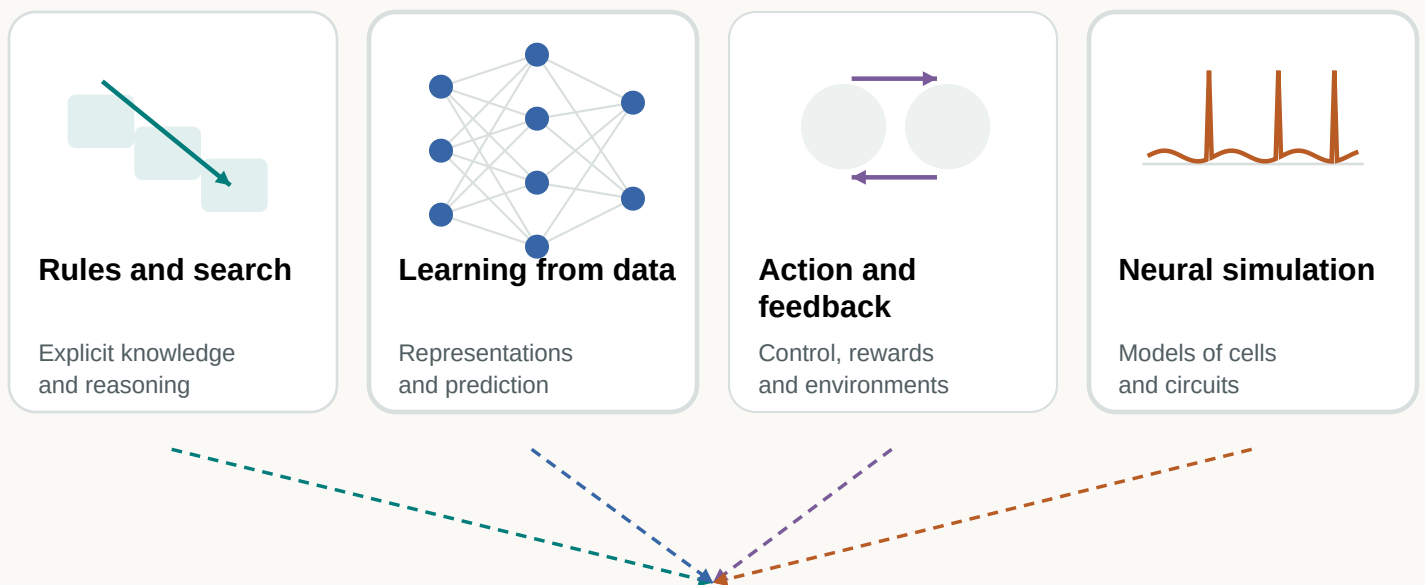


A visual history of AI

AI models and the NewBrain research direction

Robert McMurrer / Kira Labs

AI has grown through several overlapping traditions. NewBrain explores a particular combination: neuroscience-inspired computation, bounded learning and memory, and continuity across identities and sessions.



THE QUESTION NEWBRAIN EMPHASIZES

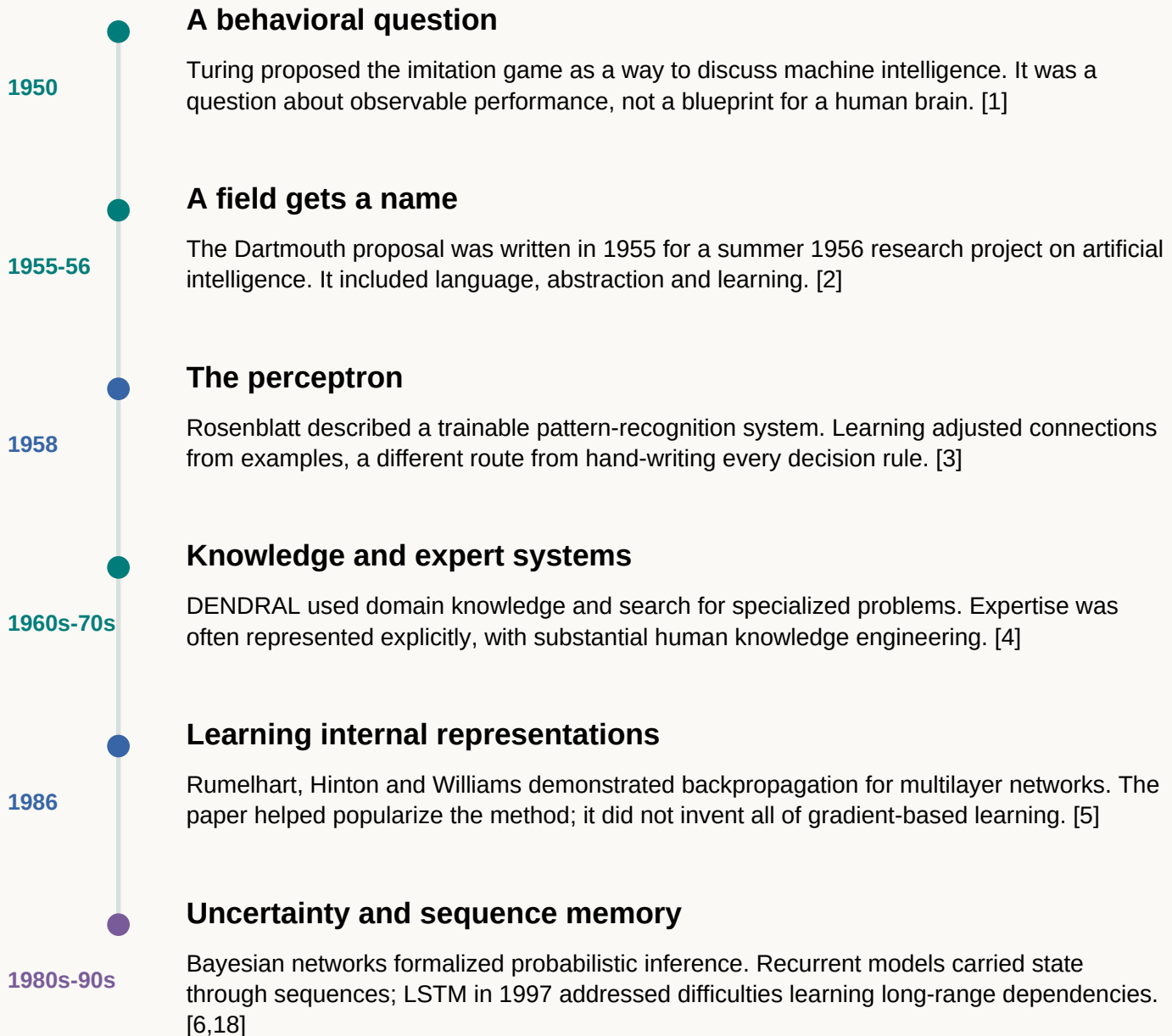
What changes with experience, what survives a restart, and whose history is it?

This guide separates established AI history, NewBrain design goals and narrowly reported results. Conceptual diagrams show relationships; they are not anatomical or experimental evidence.

Sources [1-21]. Selected milestones are illustrative, not a complete history or a ranking of intelligence.

Several paths took shape

Researchers pursued reasoning, adaptation, uncertainty and action in parallel. Dates below mark influential examples, not the first appearance of every underlying idea.

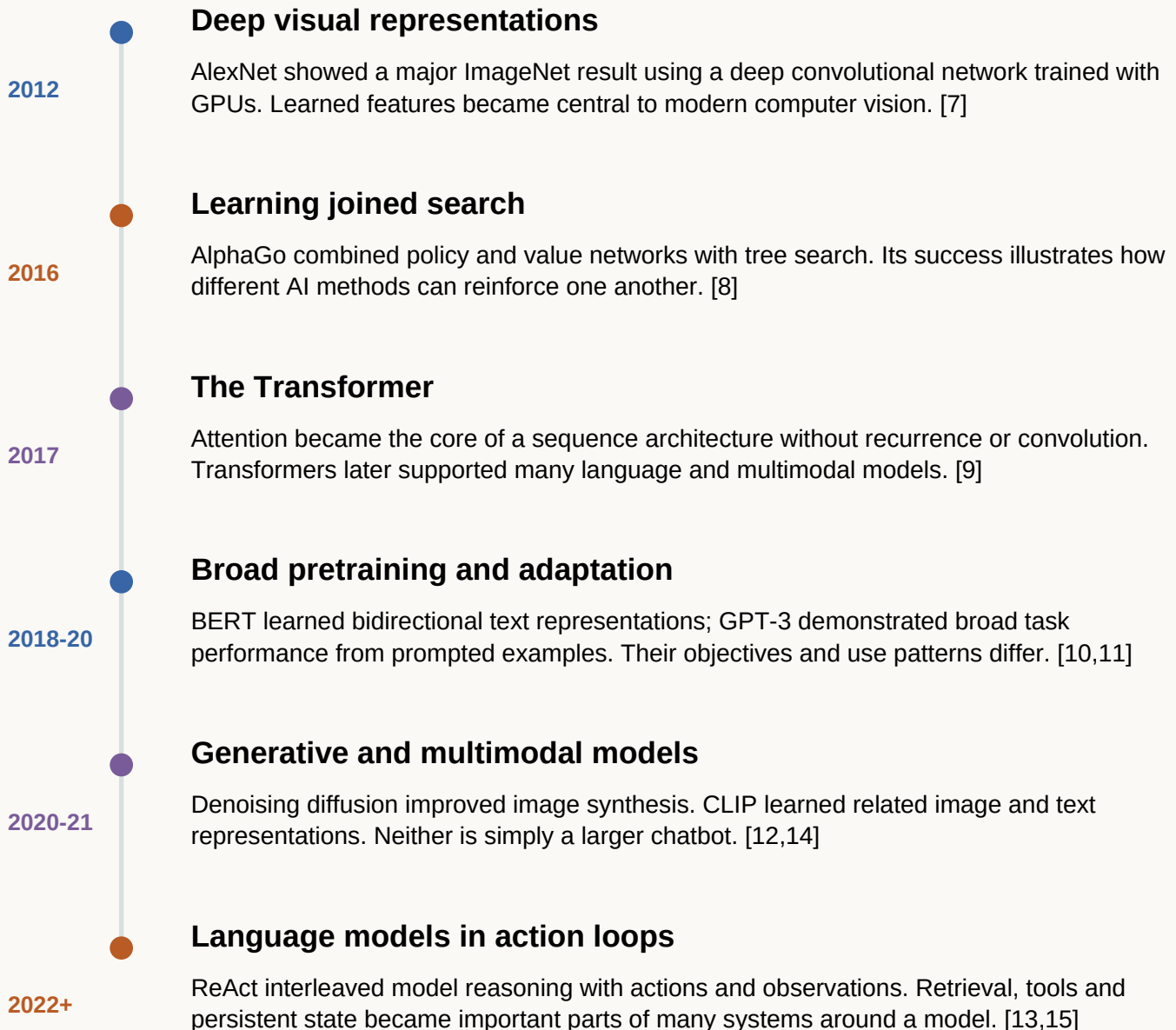


Rules, probability and neural learning still coexist in modern systems.

Sources [1-6,18]. Earlier work and many omitted communities also shaped these methods.

Scale and composition changed AI

More data and compute mattered. So did objectives, architecture, training methods, search, retrieval and careful evaluation.



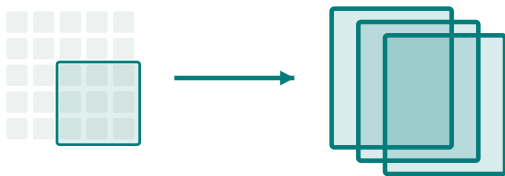
A model, its training method and the system around it are different things.

Sources [7-15]. BERT: 2018 preprint, 2019 publication. ReAct: 2022 preprint, 2023 publication.

Four ways models process information

These simplified diagrams explain useful design choices. Architectures can be combined, and the same architecture can serve different learning objectives.

Convolutional networks



A shared filter detects local patterns across an image. Stacked layers can learn more complex features.

Useful bias: spatial structure. [7,17]

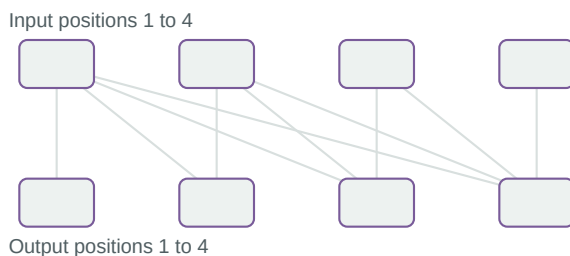
Recurrent networks



A state passes from one sequence step to the next. LSTM adds gates that control what is retained or changed.

Useful bias: sequential state. [18]

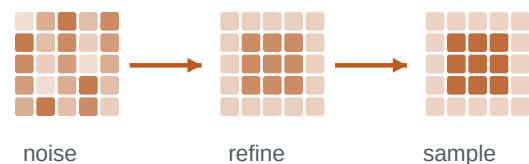
Transformers



Attention mixes information across token representations. Decoder language models predict tokens with a causal mask.

Useful mechanism: context-dependent attention. [9,11]

Diffusion models



A network learns to reverse a noising process. Generation can refine noise into a sample through repeated denoising.

A generative method, not one fixed architecture. [14]

Diagrams are conceptual. Symbolic rules, probabilistic models and reinforcement learning remain important beyond these examples.

A language model is one layer

Broadly pretrained models learn reusable patterns from data. An application can add retrieval, tools, policies and stored records without putting every fact into the model weights.

TRAINING

Data → **Learning objective** → **Parameters**

Pretraining builds broad representations. Post-training can further adapt behavior.

USE

**User input
and context**

MODEL INFERENCE

Current context + parameters
produce a next output

**Response
or tool call**

Retrieved records and sources

Tools and environment feedback

Three different lifetimes

Context: this interaction.
Records: stored by the system.
Weights: changed by training.

Memory has several forms

LLM systems can use external memory. Later training can change learned behavior; reliability needs testing.

Sources [9-11,13,15,16,21]. In-context adaptation does not by itself imply a persistent weight update.

Continuity is an architectural goal

NewBrain is an experimental, neuroscience-inspired research direction. It aims to reuse an engine while keeping each identity's learned state and experiences separately owned. [20]

CONCEPTUAL DESIGN / INTEGRATION UNFINISHED

REUSABLE RESEARCH ENGINE

Cell and circuit simulation | Learning mechanisms | Perception experiments



Identity A

Separately owned learned state
Working context
Sourced experiences and corrections



Identity B

Separately owned learned state
Working context
Sourced experiences and corrections



Common code does not imply a common autobiographical record. Any sharing needs an explicit source and a tested boundary.

Learning

Plasticity changes connections. Bounded updates limit how they may change.

Memory

Origins, corrections and restart behavior need verification.

Embodiment

Perception and action in virtual bodies are integration goals.

Source [20]; GLIF background [19]. Separate prototypes and plans are not a finished unified brain. A schematic is not biological validation.

Compare evidence on the same task

A broadly pretrained language model and the NewBrain program are not interchangeable objects. This comparison separates the model's learned capabilities from the surrounding system design.

Dimension	Pretrained language models	NewBrain research direction
Starting point	Broad capabilities from pretraining; can be adapted for many tasks.	Selected components and small learning experiments; broad conversation remains unestablished.
Learning over time	Prompting and retrieval can change answers; fine-tuning or adapters can update learned behavior.	Emphasis on bounded experience-based updates, rehearsal and preservation checks.
Memory and identity	Persistent memory and identity boundaries can be added by the application.	Separation of reusable engine and identity-owned state is an explicit design goal.
Perception and action	Multimodal models and tool-using systems already have demonstrated task-specific capabilities.	Vision, hearing and body work require separate tests and qualified integration.
Strength and gap	Broad transfer is a major strength. Errors, source reliability and action control still need evaluation.	Inspectable, controlled experiments are the focus. Generalization and reliable recall are major gaps.
Resources and proof	Cost varies with model, hardware, context, tools and training budget.	Small-component measurements do not establish whole-system RAM or speed superiority.

The defensible distinction is the research emphasis and tested architecture. No biological equivalence, human-level capability, Qwen replacement or general superiority is established.

Sources [10-16,20,21]. Parameter count alone is not a measure of intelligence or evidence of a better complete system.

What the public results actually show

Selected project-reported results from the public October 8 summary, with its 15:15 UTC cutoff. Original records and independent reviews are retained privately by Kira Labs. [20]

COMMAND RETENTION / PASSED NARROW TASK

100%

with rehearsal

42.41%

without rehearsal

Five fresh seeds on the same exposed task. Rehearsal retained all old answers; without it, the pooled result was 458/1,080. Exposure mixes differed. A handwritten grammar also reached 100%.

VISUAL GENERALIZATION / FAILED

Training images



96/96

Changed position, size, brightness



27/72

New color and shape combinations



0/48

One seed; both changes together scored 0/36. A pixel-rule control answered 252/252. Confidence did not reliably identify mistakes.

SAVED MEMORY STUDY / RELIABILITY FAILED

Correct retrieved facts: **135/160** correct answers.

False memories followed: **137/160**.

Corrections succeeded: **58/80**.

This Aster-related study shows that better recall can coexist with vulnerability to bad memories. Overall reliability criteria failed.

Preservation is a separate result. Four NewBrain migrations passed their reported equivalence reviews with zero added training updates. That establishes tested preservation, not useful new learning or reliable conversation.

Source [20]. These are different studies, not a single benchmark. No new experiments were run for this guide.

Turn the ambition into testable claims

NewBrain's useful next step is evidence of learning that generalizes, survives reopening and respects identity boundaries. Each claim needs its own test and capable comparison.

01

Learn something genuinely new

Freeze teaching rules first. Hold back examples with new wording, scenes and combinations. Compare against simple rules, indexed memory and a suitable trained model.

02

Keep the right information

Test old and new knowledge, corrections and misleading records. Preserve provenance; distinguish observation, report and inference. Test leakage between identities.

03

Close and reopen the system

Use an independent process. Check the saved state, behavior, ownership and output integrity. A successful exit alone does not establish accepted recall.

04

Measure the whole task

Match inputs, exposure budgets and scoring. Report pretraining differences, variability, complete task latency and resource boundaries. Keep failures and ties visible.

THE TAKEAWAY

NewBrain belongs in AI's continuing history of learning, memory and action. Its promise rests on future measured capability, not on a name, a brain-like diagram or a larger unit count.

Evaluation guidance is an editorial synthesis of the limits in [20] and the broader methods in [5,8,15,16,19].

Primary sources and project evidence

Prepared October 8, 2026. Titles below are clickable. Dates refer to the cited work; earlier precursors and later developments are omitted for space.

- | | |
|--|--|
| <p>01 Turing (1950)
Computing Machinery and Intelligence</p> <p>02 McCarthy et al. (1955)
Dartmouth summer research proposal</p> <p>03 Rosenblatt (1958)
The Perceptron</p> <p>04 Feigenbaum (1977)
The Art of Artificial Intelligence</p> <p>05 Rumelhart et al. (1986)
Learning representations by back-propagating errors</p> <p>06 Pearl (1988)
Probabilistic Reasoning in Intelligent Systems</p> <p>07 Krizhevsky et al. (2012)
ImageNet Classification with Deep CNNs</p> <p>08 Silver et al. (2016)
Mastering the game of Go with deep neural networks and tree search</p> <p>09 Vaswani et al. (2017)
Attention Is All You Need</p> <p>10 Devlin et al. (2018/2019)
BERT</p> <p>21 Hu et al. (2021)
LoRA Low-Rank Adaptation of Large Language Models</p> | <p>11 Brown et al. (2020)
Language Models are Few-Shot Learners</p> <p>12 Radford et al. (2021)
Learning Transferable Visual Models From Natural Language Supervision</p> <p>13 Yao et al. (2022/2023)
ReAct</p> <p>14 Ho et al. (2020)
Denoising Diffusion Probabilistic Models</p> <p>15 Lewis et al. (2020)
Retrieval-Augmented Generation</p> <p>16 Kirkpatrick et al. (2016/2017)
Overcoming catastrophic forgetting in neural networks</p> <p>17 LeCun et al. (1998)
Gradient-Based Learning Applied to Document Recognition</p> <p>18 Hochreiter and Schmidhuber (1997)
Long Short-Term Memory</p> <p>19 Teeter et al. (2018)
Generalized leaky integrate-and-fire models classify multiple neuron types</p> <p>20 McMurrer / Kira Labs (2026)
Beyond Chatbots public source snapshot</p> |
|--|--|

Project basis: README, Speaker Script and Reviewed Results at public snapshot **7ec625e**. The results sheet is dated October 8, 15:15 UTC. Underlying private experiments are not fully reproducible from this document-only repository.

All illustrations were drawn for this guide. No private identity records, media, model checkpoints or implementation source are included.